

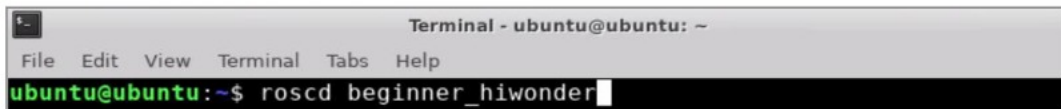
第 7 课 话题消息的自定义与使用

1.自定义话题消息

注意：在自定义话题消息之前，需要先创建工作空间和功能包，具体操作过程可前往目录“第2章 ROS开发入门->ROS基础课程->第3课 创建工作空间与功能包”查看文档。

自定义话题消息的具体操作步骤如下：

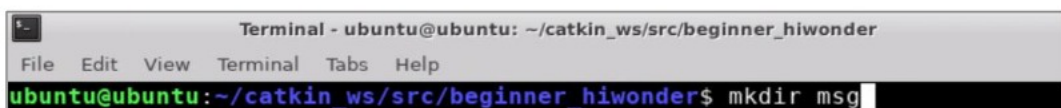
- 1) 打开命令行终端。
- 2) 输入进入功能包所在目录的指令“`roscd beginner_hiwonder`”，并按下回车。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscd beginner_hiwonder
```

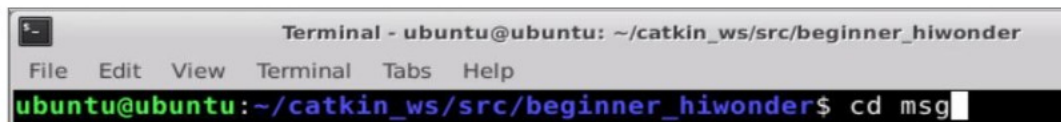
注意：若出现提示“`No such package/stack 'beginner_hiwonder'`”，即功能包不存在于环境变量`ROS_PACKAGE_PATH`中，具体解决方法可前往目录“第2章 ROS开发入门->ROS基础课程->第3课 创建工作空间与功能包”查看文档，解决此问题后，请重复一次当前步骤的操作。

- 3) 输入指令“`mkdir msg`”，并按下回车，新建用于存放文本文件的文件夹“`msg`”。



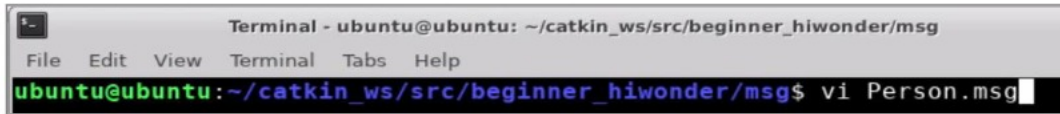
```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ mkdir msg
```

- 4) 输入指令“`cd msg`”，回车。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ cd msg
```

5) 输入指令“**vi Person.msg**”编辑程序，复制下面程序。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/msg
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/msg$ vi Person.msg
```

```
string name

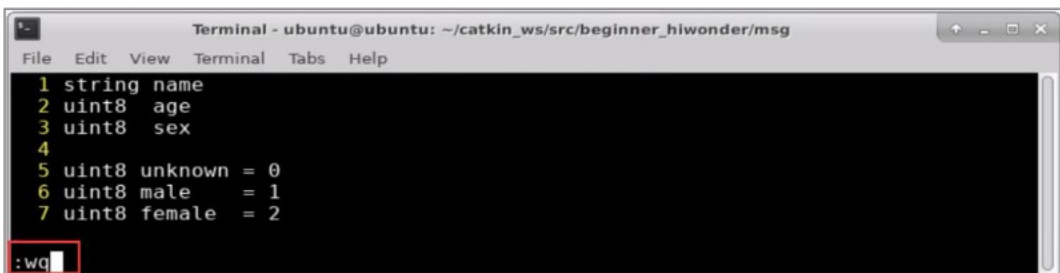
uint8 age

uint8 sex

uint8 unknown = 0

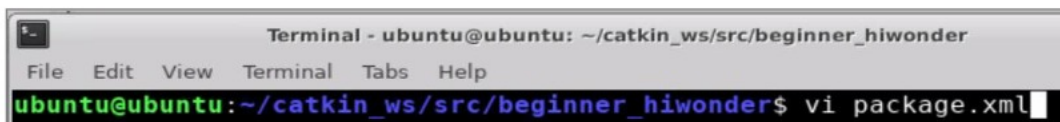
uint8 male    = 1

uint8 female  = 2
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/msg
File Edit View Terminal Tabs Help
1 string name
2 uint8 age
3 uint8 sex
4
5 uint8 unknown = 0
6 uint8 male    = 1
7 uint8 female  = 2
:wq
```

6) 输入“**vi package.xml**”，复制下面程序，在下图所示位置添加功能包依赖。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ vi package.xml
```

7) 在下图所示位置添加功能包依赖：

```
<build_depend>message_generation</build_depend>

<exec_depend>message_runtime</exec_depend>
```

```

49 <!-- Use doc_depend for packages you need only for building documentation:
-->
50 <!-- <doc_depend>doxygen</doc_depend> -->
51 <buildtool_depend>catkin</buildtool_depend>
52 <build_depend>roscpp</build_depend>
53 <build_depend>rospy</build_depend>
54 <build_depend>std_msgs</build_depend>
55 <build_export_depend>roscpp</build_export_depend>
56 <build_export_depend>rospy</build_export_depend>
57 <build_export_depend>std_msgs</build_export_depend>
58 <exec_depend>roscpp</exec_depend>
59 <exec_depend>rospy</exec_depend>
60 <exec_depend>std_msgs</exec_depend>
61 <build_depend>message_generation</build_depend>
62 <exec_depend>message_runtime</exec_depend>
63

```

- 8) 输入“vi CMakeLists.txt”，再按下“i”，修改“CMakeLists.txt”文件。

```

Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ vi CMakeLists.txt

```

- 9) 在下图所示位置添加所需的编译选项“message_generation”。

```

7 ## Find catkin macros and libraries
8 ## if COMPONENTS list like find_package(catkin REQUIRED COMPONENTS xyz)
9 ## is used, also find other catkin packages
10 find_package(catkin REQUIRED COMPONENTS
11   roscpp
12   rospy
13   std_msgs
14   message_generation
15 )

```

- 10) 找到下图所示代码，将红框部分进行反注释，并添加所需的编译选项“Person.msg”。

```

49
50 ## Generate messages in the 'msg' folder
51 # add_message_files(
52 #   FILES
53 #   Message1.msg
54 #   Message2.msg
55 # )
56

```

```

50 ## Generate messages in the 'msg' folder
51 add_message_files(
52   FILES
53   Person.msg
54 )
55

```

11) 找到下图所示代码，将红框部分进行反注释，确保所需的编译选项生效。

```
70 ## Generate added messages and services with any dependencies listed here
71 # generate_messages(
72 #   DEPENDENCIES
73 #     std_msgs
74 # )
75
```

```
70 ## Generate added messages and services with any dependencies listed here
71 generate_messages(
72   DEPENDENCIES
73     std_msgs
74 )
75
```

12) 找到下图所示代码，将红框部分进行反注释，并添加所需的编译选项“message_runtime”。

```
105 catkin_package(
106 #   INCLUDE_DIRS include
107 #   LIBRARIES beginner_hiwonder
108 #   CATKIN_DEPENDS roscpp rospy std_msgs
109 #   DEPENDS system_lib
110 )
```

```
105 catkin_package(
106 #   INCLUDE_DIRS include
107 #   LIBRARIES beginner_hiwonder
108 CATKIN_DEPENDS roscpp rospy std_msgs message_runtime
109 #   DEPENDS system_lib
110 )
```

13) 修改完成，按下“Esc”，输入“:wq”保存并退出。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
184 #   FILES_MATCHING PATTERN "*.h"
185 #   PATTERN ".svn" EXCLUDE
186 # )
:wq
```

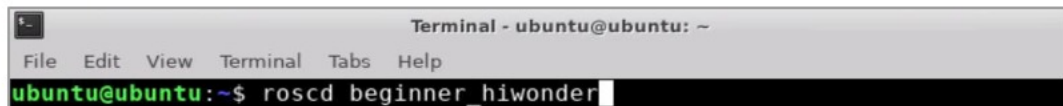
14) 输入指令“rosmmsg show beginner_hiwonder/Person”，并按下回车，查看写入的消息字段能否被系统识别。当出现下图红框所示字样，即代表识别成功。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ rosmmsg show beginner_hiwonder/Person
uint8 unknown=0
uint8 male=1
uint8 female=2
string name
uint8 age
uint8 sex
```

2.话题消息的使用

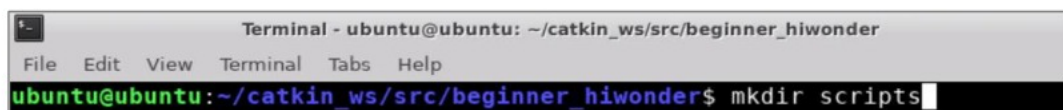
2.1 创建发布者和订阅者代码

- 1) 打开命令行终端。
- 2) 输入进入功能包所在目录的指令“**roscd beginner_hiwonder**”，并按下回车。



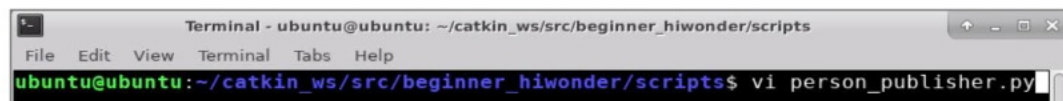
```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscd beginner_hiwonder
```

- 3) 输入指令“**cd scripts**”，并按下回车，进入用于存放Python脚本的文件夹“**scripts**”。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ mkdir scripts
```

- 4) 输入指令“**vi person_publisher.py**”编辑程序，复制下面程序。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi person_publisher.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程将发布/person_info话题，自定义消息类型beginner_hiwonder::Person

import rospy

from beginner_hiwonder.msg import Person

def velocity_publisher():
```



```
# ROS节点初始化
```

```
rospy.init_node('person_publisher', anonymous=True)
```

```
# 创建一个 Publisher，发布名为 /person_info 的 topic，消息类型为  
beginner_hiwonder::Person，队列长度10
```

```
person_info_pub = rospy.Publisher('/person_info', Person, queue_size=10)
```

```
#设置循环的频率
```

```
rate = rospy.Rate(10)
```

```
while not rospy.is_shutdown():
```

```
    # 初始化beginner_hiwonder::Person类型的消息
```

```
    person_msg = Person()
```

```
    person_msg.name = "Tom";
```

```
    person_msg.age  = 18;
```

```
    person_msg.sex  = Person.male;
```

```
    # 发布消息
```

```
    person_info_pub.publish(person_msg)
```

```
    rospy.loginfo("Publsh person message[%s, %d, %d]",
```

```
                    person_msg.name, person_msg.age, person_msg.sex)
```

```
# 按照循环频率延时

rate.sleep()

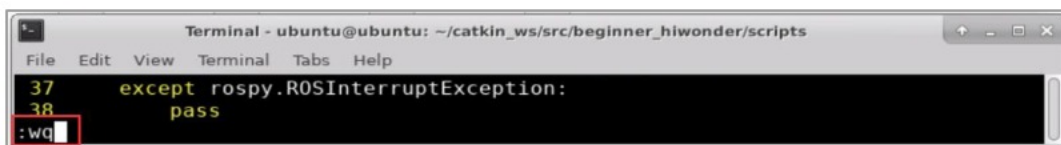
if __name__ == '__main__':

    try:

        velocity_publisher()

    except rospy.ROSInterruptException:

        pass
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
37     except rospy.ROSInterruptException:
38     pass
:wq
```

5) 输入指令“**vi person_subscriber.py**”编辑程序，复制下面程序。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi person_subscriber.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程将订阅/person_info话题，自定义消息类型beginner_hiwonder::Person

import rospy

from beginner_hiwonder.msg import Person
```

```

def personInfoCallback(msg):

    rospy.loginfo("Subscribe Person Info: name:%s  age:%d  sex:%d",

                  msg.name, msg.age, msg.sex)

def person_subscriber():

    # ROS节点初始化

    rospy.init_node('person_subscriber', anonymous=True)

    # 创建一个 Subscriber，订阅名为 /person_info 的 topic，注册回调函数
personInfoCallback

    rospy.Subscriber("/person_info", Person, personInfoCallback)

    # 循环等待回调函数

    rospy.spin()

if __name__ == '__main__':

    person_subscriber()

```



```

Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
1 #!/usr/bin/env python
2 # -*- coding: utf-8 -*-
3
4 # 该例程将订阅/person_info话题，自定义消息类型beginner_hiwonder::Person
5
6 import rospy
7 from beginner_hiwonder.msg import Person
:wq

```

- 6) 输入指令“**chmod +x person_publisher.py**”回车，为保存的person_publisher.py赋

予可执行权限。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x person_publisher.py
```

7) 输入指令“`chmod +x person_subscriber.py`”回车，为保存的`person_publisher.py`赋予可执行权限。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x person_subscriber.py
```

2.2 运行发布者和订阅者节点

1) 输入指令“`cd ~/catkin_ws`”，并按下回车，进入`catkin`工作空间。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ cd ~/catkin_ws
```

2) 输入指令“`catkin_make`”，并按下回车，构建目录中所有的功能包。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ catkin_make
```

3) 输入指令“`source ./devel/setup.bash`”，并按下回车，刷新工作空间的环境。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ source ./devel/setup.bash
```

4) 输入指令“`roscore`”，启动节点管理器。

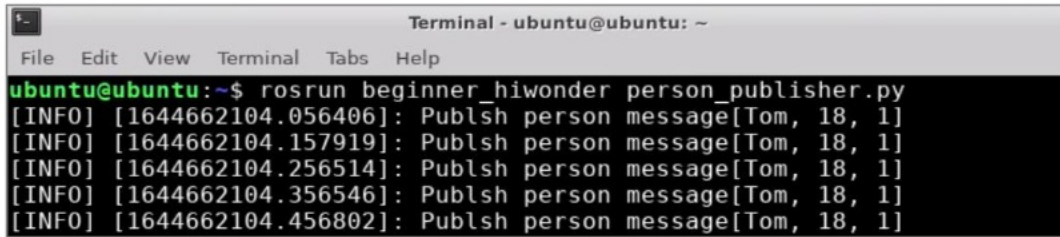
```
ubuntu@ubuntu:~/catkin_ws$ roscore
```

若已开启，则会出现以下提示：

```
RLException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://ubuntu:11311/
The traceback for the exception was written to the log file
```

5) 输入指令“`roslaunch beginner_hiwonder person_publisher.py`”，并按下回车，运行

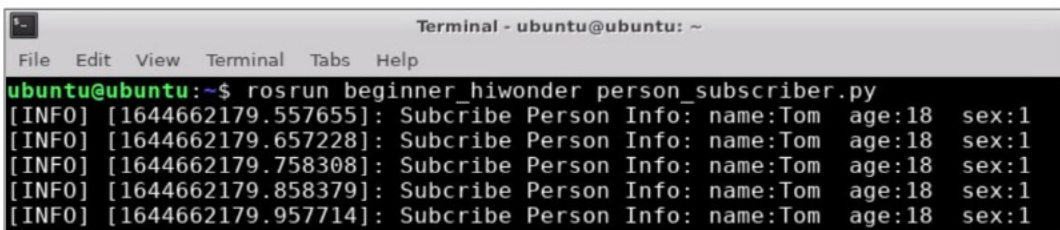
发布者节点。如需停止节点的运行，可按下快捷键“Ctrl+C”。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosrn beginner_hiwonder person_publisher.py
[INFO] [1644662104.056406]: Publsn person message[Tom, 18, 1]
[INFO] [1644662104.157919]: Publsn person message[Tom, 18, 1]
[INFO] [1644662104.256514]: Publsn person message[Tom, 18, 1]
[INFO] [1644662104.356546]: Publsn person message[Tom, 18, 1]
[INFO] [1644662104.456802]: Publsn person message[Tom, 18, 1]
```

6) 打开一个新的命令行终端，输入指令

“`rosrn beginner_hiwonder person_subscriber.py`”，并按下回车，运行订阅者节点。如需停止节点的运行，可按下快捷键“Ctrl+C”。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosrn beginner_hiwonder person_subscriber.py
[INFO] [1644662179.557655]: Subcribe Person Info: name:Tom age:18 sex:1
[INFO] [1644662179.657228]: Subcribe Person Info: name:Tom age:18 sex:1
[INFO] [1644662179.758308]: Subcribe Person Info: name:Tom age:18 sex:1
[INFO] [1644662179.858379]: Subcribe Person Info: name:Tom age:18 sex:1
[INFO] [1644662179.957714]: Subcribe Person Info: name:Tom age:18 sex:1
```

注意：

- 1) 需要先启动发布者节点，订阅者节点才可订阅消息；
 - 2) 若需完全接收发布者消息，可以先启动订阅者节点，再启动发布者节点。
-